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EE 3001: Solid-State Devices

Quiz 2

Put down your answers in the respective places within the question paper. The steps to get the answer must be shown in the additional sheets.
All numerical answers should have proper units.

Date: 26.03.2025 Time: 50 minutes Marks: 20

1. Fig.1 shows the electric field variation near the abrupt junction of a p-type and an n-type semiconductor located at x = 0. The doping concentration on the p-side is given by $N_A = 10^{16}/cm^3$ and built-in potential is $V_{bi} = 0.75$ V.

(a) **The doping concentration on the n-side (N_D) is** $4\times10^{16}/cm^3$ X [2]

(b) **The applied voltage at p-side with respect to n-side is** -1.125 V [2] (2)

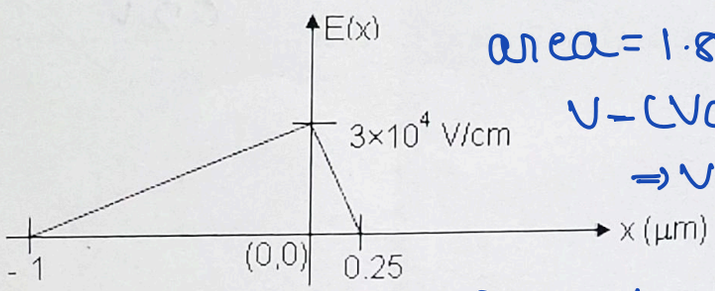


Fig.1

area = 1.875V

$V-(V_{app}) = 1.875$

$\Rightarrow V_{app} = -1.125$

$I_p = \frac{Aqn_i^2 D_p}{L_p N_D}(e^{V/V_T}-1)$ $I_n = \frac{Aqn_i^2 D_n}{L_n N_A}(e^{V/V_T}-1)$ $I_n = 8I_p$

2. In an abrupt p-n junction with long p-region and n-region of uniform cross-sectional area, the doping concentrations in the p-region and n-region are given by $N_A = 1 \times 10^{17}/cm^3$ and $N_D = 4 \times 10^{17}/cm^3$, respectively. Assume that electron mobility (μ_n) is four times hole mobility (μ_p), i.e. $\mu_n = 4\mu_p$ and the minority carrier lifetime for electrons (τ_n) is equal to the minority carrier lifetime for holes (τ_p), i.e. $\tau_n = \tau_p$. The current flowing through the forward biased device is **6 mA**. Assume net generation/recombination current in the depletion region to be zero.

$I_n + I_p = 6mA$

(a) **The electron current (I_n) at the edge of the depletion region is** 5.33333333 mA [2] 16/3mA

(b) **The hole current (I_p) at the edge of the depletion region is** 0.66666666 mA [2] 2/3mA (4)

3. An abrupt p^+n diode has a uniform cross-sectional area (A) equal to 1 cm^2. When the applied forward bias (V_f) is 0.6 V, the width of the neutral n-region (W_n) is 2 μm and is much less than the hole diffusion length (L_p) in this region and the hole concentration at the edge of the depletion region in the n-region is $1 \times 10^{11}/cm^3$. The hole diffusion constant (D_p) in this region is 10 cm^2/sec. **Assume $V_T = kT/q = 25$ mV.**

(a) **The small signal conductance (g) at $V_f = 0.6$ V is** $0.032(\Omega m)^{-1}$ X [2]

(b) **The diffusion capacitance (C_d) at $V_f = 0.6$ V is** 6.4 nF 64pF [2]

$I = I_0(e^{V/V_T}-1)$ $I_0 = \frac{qAD_p p_{no}}{W_n}$ $g = \frac{dI}{dV} = \frac{I}{V_T} = \frac{0.6mA}{25mV}$

$C_d = g.\tau_T = \frac{W_n^2}{2D_p} \times G$

$C_j = A\sqrt{\frac{q\varepsilon_{Si} N_D}{2(V_{bi}+V_R)}}$ $\quad 4 = \frac{V_{bi}+12}{V_{bi}+2.4} \Rightarrow V_{bi} = 0.8$

4. At T = 300 K, the depletion capacitance values of a p^+n junction silicon diode with cross-sectional area of 1 mm^2 were measured to be 50 pF and 25 pF at applied reverse bias voltages of 2.4 V and 12 V respectively.
Consider the permittivity of silicon to be 10^{-12} F/cm.

(a) **Depletion capacitance value at zero bias will be** ..... 100 PF ✓ [2]

(b) **Built-in potential value of the diode is** ..... 0.8 V ✓ ..... [2]

(c) **Doping concentration of the n-region is** ..... $10^{15}/cm^3$ ✓ ..... [2]

At a higher temperature (T = T_H), the depletion capacitance value of the same diode at zero bias was found to be 200 pF.

(d) **Built-in potential of the diode at** T = T_H **is** ..... 0.2V ✓ ..... [2]

8